The zero-sum sheath stabiliser, spelled out

from SOLPS-ITER `b2stbc_stab_coeff_sheath_te/ti/ni`

Step 1 — the equation. The wall temperature is heated at rate Q and drained by the sheath at rate kT:

$$\frac{\mathrm{d}T}{\mathrm{d}t} = Q - kT \qquad \Longrightarrow \qquad \text{equilibrium when nothing changes:} \quad 0 = Q - kT^* \quad \Rightarrow \quad \boxed{T^* = Q/k}$$

Step 2 — one implicit time step. T^{old} = the value we have; T^{new} = the value we solve for. "Implicit" means the sink uses T^{new}:

$$\frac{T^{\text{new}} - T^{\text{old}}}{\Delta t} = Q - k\,T^{\text{new}} \qquad \Longrightarrow \qquad T^{\text{new}} = \frac{T^{\text{old}} + Q\Delta t}{1 + k\Delta t}$$

Step 3 — add the stabiliser. We add one extra term to the source:

$$S^{\text{stab}} = \alpha k \big(T^{\text{old}} - T^{\text{new}}\big)$$

Read it carefully: it is the difference between the old and the new value. While T is still changing it is non-zero and resists the change. Once T settles, $T^{\text{new}} = T^{\text{old}}$ and the term is **exactly zero** — it switches itself off.

Putting it in and collecting T^{new} on the left:

$$\frac{T^{\text{new}} - T^{\text{old}}}{\Delta t} = Q - kT^{\text{new}} + \alpha k(T^{\text{old}} - T^{\text{new}}) \qquad \Longrightarrow \qquad T^{\text{new}} = \frac{T^{\text{old}}\,(1 + \alpha k\Delta t) + Q\Delta t}{1 + k\Delta t + \alpha k\Delta t}$$

Step 4 — does it move the answer? Set $T^{\text{new}} = T^{\text{old}} = T^*$ and see.

$$T^*(1 + k\Delta t + \alpha k\Delta t) = T^*(1 + \alpha k\Delta t) + Q\Delta t$$

$$T^* + k\Delta t\,T^* + \underbrace{\alpha k\Delta t\,T^*}_{\text{cancels}} = T^* + \underbrace{\alpha k\Delta t\,T^*}_{\text{cancels}} + Q\Delta t \qquad \Longrightarrow \qquad k\Delta t\,T^* = Q\Delta t$$

$\boxed{T^* = Q/k}$ **every α term cancelled — the answer is unchanged.**

Step 5 — so what *does* it change? The step size. Same equation, real numbers ($Q = 5$, $k = 2$, $\Delta t = 0.5$, start at $T = 9$, so $T^* = 2.5$):

	start	after step …						after 4000
		1	2	3	4	5	6	
$\alpha = 0$	9.000	5.750	4.125	3.313	2.906	2.703	2.602	**2.5000000000**
$\alpha = 100$	9.000	8.936	8.873	8.811	8.749	8.688	8.627	**2.5000000000**

- Both columns end at **exactly the same** $T^* = 2.5$.
- $\alpha = 100$ simply takes *smaller bites*: no single step can overshoot.
- That is the entire mechanism. **Same destination, smaller steps.**

Step 6 — why this is not just "adding damping to make it stable". In an implicit solve there are two separate objects:

residual	"how far am I from balance?" — the answer is where this is **zero**
Jacobian	"how big a step do I take?" — **does not appear** in that condition

S^{stab} is zero at balance, so it adds **nothing** to the residual; its slope $-\alpha k$ is not zero, so it adds to the Jacobian. It changes only the second column. In the code this is literally: $(\gamma_{\text{sheath}} - 1 + \alpha)$ in the **Jacobian**, $(\gamma_{\text{sheath}} - 1)$ in the **residual**.

Careful: if we had changed *both*, we would simply have increased γ_{sheath} — a different sheath, a different answer. Only the Jacobian is touched.

Step 7 — and α is not a free fudge factor. It is dimensionless, a multiple of the sheath transmission coefficient already in the term. SOLPS ships it, recommends 1–100, defaults to 0; $\alpha = 0$ reproduces our earlier results bit-for-bit. It damps the **transport** rows (n_e, T_i, T_e) — SOLPS ships none for the momentum condition — which matches our failure, where n_e and T_e went negative *together* in the same 2–3 cells.

Result: drift-enabled run previously died at ≤ 649 steps $\rightarrow$ **now past 1100 and converging.**

Two honest notes. (i) The scalar example above demonstrates the *property* — same fixed point, smaller steps — not the instability itself, which comes from the coupling between equations. (ii) SOLPS iterates inside a time step, so there the term vanishes before the step is accepted and is purely a convergence aid; JOREK takes one solve per step, so T^{old} is the previous *time step* and real transients are slowed too. The stationary state is unaffected either way.